CHAPTER – 12 – ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

LECTURE NOTES

FOR CBSE / IIT – JEE / NEET EXAMINATIONS

2023 – 2024

Unit XII: ALDEHYDES KETONES AND CARBOXYLIC ACIDS

SYLLABUS FOR 2023 – 2024

ALDEHYDES & KETONES

Nomenclature,

Nature of carbonyl group,

Methods of preparation,

Physical and chemical properties,

Mechanism of nucleophilic addition,

Reactivity of alpha hydrogen in aldehydes;

Uses

CARBOXYLIC ACIDS:

Nomenclature,

Acidic nature,

Methods of preparation,

Physical and chemical properties;

Uses

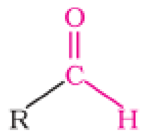
Weightage of This Chapter for CBSE Board Examination = 8 Marks

Carbonyl Compounds

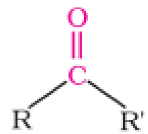
Carbonyl Compounds are the organic compounds containing carbon-oxygen double bond (>C=0).

Carbonyl compounds in which carbonyl group is bonded to a carbon and hydrogen are known as aldehydes.

Carbonyl compounds in which carbonyl group is bonded to carbon atoms are known as ketones.



Aldehyde



Ketone

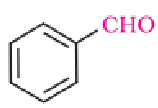
Nomenclature of Aldehydes and Ketones

(a) Common names

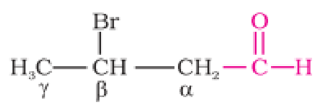
The common name of aldehydes are derived from the names of the corresponding carboxylic acids by replacing –ic acid by –aldehyde.

CH_3CHO

Acetaldehyde

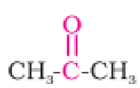


Benzaldehyde

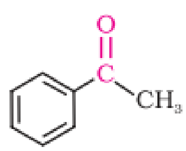


β-Bromobutyraldehyde

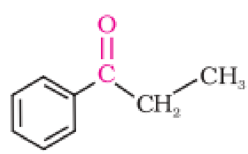
The simple aliphatic ketone has the common name acetone. For most other aliphatic ketones we name the two groups that are attached to carbonyl carbon and follow these names by the word ketone. A ketone in which the carbonyl group is attached to a benzene ring is named as a phenone, all illustrated below.



Acetone



Acetophenone



Propiophenone

(b) IUPAC names

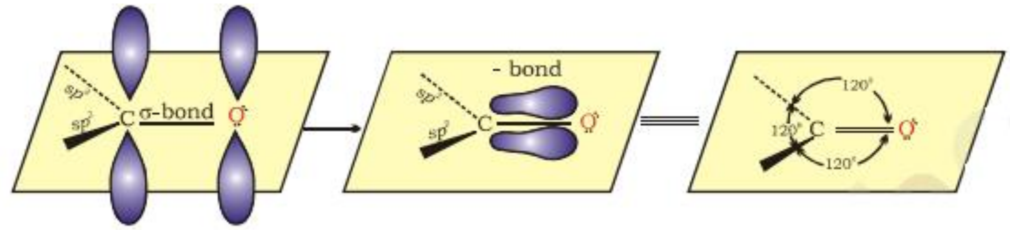
(i) The longest chain containing the –CHO group is considered the parent structure and named by replacing –e of the corresponding alkane by –al.

(ii) The position of the substituent is indicated by a number, the carbonyl carbon always being considered C-1.

(iii) According to IUPAC system, the longest chain carrying the carbonyl group is considered the parent structure, and is named by replacing –e of the corresponding alkane with one. The positions of various groups are indicated by numbers.

Structure of Carbonyl Group

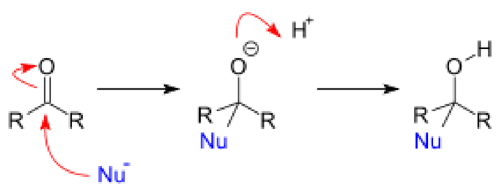
Carbonyl carbon is joined to three other atoms by s bonds; since these bonds utilize sp[2] orbitals, they lie in a plane, and are 120° apart. The remaining p-orbitals of carbon overlaps a p-orbital of oxygen to form a pi bond; carbon and oxygen are thus joined by a double bond. The part of the molecule immediately surrounding carbonyl carbon lie in a plane.



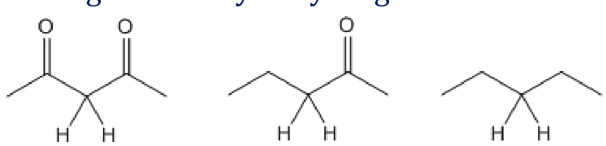
The electrons of a carbonyl double bond hold together atoms of quite different electronegativity and hence the electrons are not equally shared; in particular, the polar p-cloud is pulled strongly towards the more electronegative atom, oxygen.

The carbonyl group, C = O, governs the chemistry of aldehydes and ketones in two ways:

a) By providing a site for nucleophilic addition, and



b) By increasing the acidity of hydrogen atoms attached to the alpha carbon.



A diketone pKa 9 | A ketone pKa 19 | A hydrocarbon pKa 51

Preparation of Aldehydes and Ketones

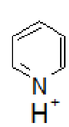
- **Preparation of Aldehydes and Ketones**

1. By oxidation of alcohols

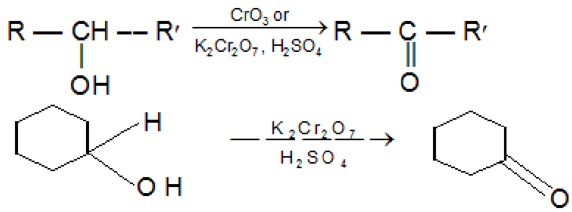
(i) Oxidation of primary alcohols (By Special Distillation Method)

$$\underset{\text{1° alcohol}}{RCH_2OH} \xrightarrow{K_2Cr_2O_7} \underset{\text{Aldehyde}}{R-\overset{\overset{H}{|}}{C}=O}$$

$$RCH_2OH \xrightarrow[CH_3Cl]{P.C.C.} R-CHO$$

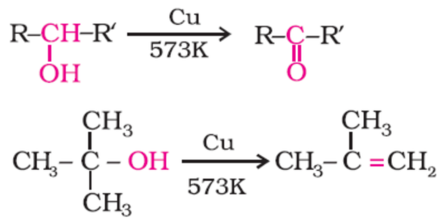
 $CrO_3Cl \longrightarrow$ (Pyridinium chloro-chromate) (P.C.C.)

(ii) Oxidation of Secondary alcohols



2. By dehydrogenation of alcohols

$$RCH_2OH \xrightarrow[573K]{Cu} RCHO$$



3. From hydrocarbons

(i) By ozonolysis of alkenes

Ozone reacts vigorously with alkenes to form unstable compounds called ozonides. Ozonides, themselves are unstable and reduced directly with Zn and water. The reduction produces carbonyl compounds that can be isolated and identified.

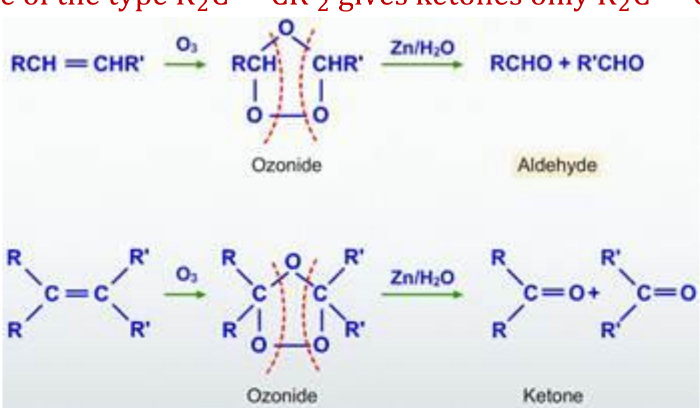
The ozonolysis thus involves the replacement of an olefinic bond C=C by two carbonyl groups C=O. The total number of carbon atoms in two carbonyl compound is equal to total number of carbon atoms in alkene.

The Zn dust used during hydrolysis checks the formation of H_2O_2 which can otherwise oxidize the products (carbonyl compounds) to respective acids.

An alkene of the type $RCH = CHR'$ gives two aldehydes $RCHO + R'CHO$

An alkene of the type $R_2C = CHR'$ gives $R_2C = O + R'CHO$

An alkene of the type $R_2C = CR'_2$ gives ketones only $R_2C = O + R'_2C = O$



Reduction of ozonide can also be made by Zn/Acid, H_2-Raney Ni or triphenyl phosphine to carbonyl compounds.

Reduction of ozonide by $LiAlH_4$ or sodium borohydride gives corresponding alcohols.

(ii) By hydration of alkynes (Hydration to Carbonyl Compounds):

When passed into dilute sulphuric acid at 60°C in the presence of mercuric sulphate as catalyst, acetylene adds on one molecule of water to form acetaldehyde.

The mechanism of this hydration takes place via the formation of vinyl alcohol as an intermediate.

$$HC \equiv CH + H_2O \xrightarrow{\text{dil. } H_2SO_4,\ Hg^{+2}} [CH_2 = CH(OH)] \longrightarrow CH_3CHO$$

Vinyl alcohol $CH_2 = CH(OH)$, which is rapidly converted into an equilibrium mixture that is almost CH_3CHO is an example of keto-enol tautomerism.

The homologues of acetylene form ketone when hydrated, example, propyne gives acetone.

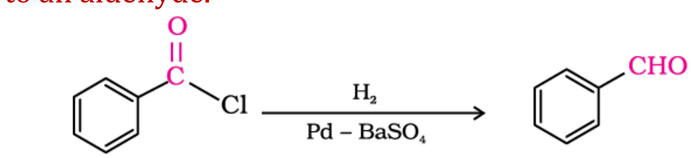
$$CH_3 - C \equiv CH + H_2O \xrightarrow{\text{dil. } H_2SO_4,\ Hg^{+2}} CH_3 - CO - CH_3$$

Only C_2H_2 on addition of H_2O gives aldehyde and rest all alkynes give ketone.

- **Preparation of Aldehydes**

1. Reduction of acid chlorides (Rosenmund Reduction)

The Rosenmund reduction is a hydrogenation process in which an acyl chloride is selectively reduced to an aldehyde.



The Pd catalyst must be poisoned, for example with $BaSO_4$, because the untreated catalyst is too reactive and will give some over reduction (Aldehyde formed, can further reduced to alcohol).

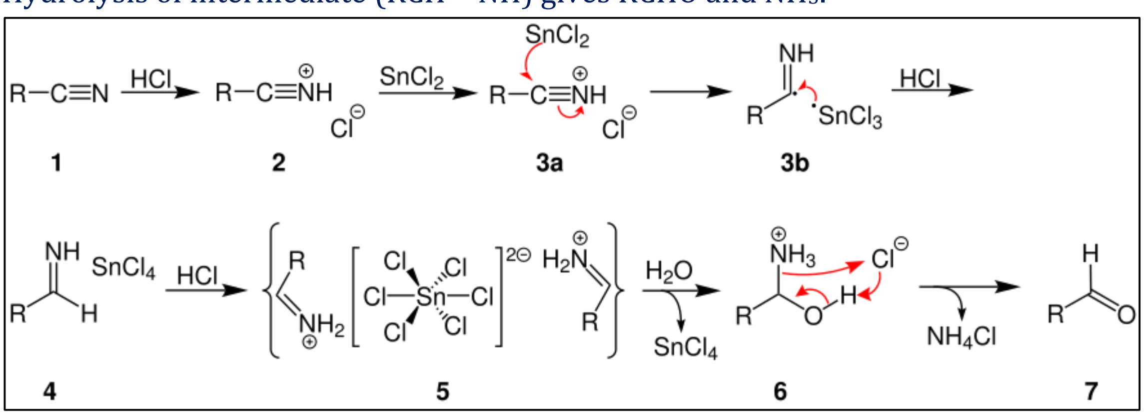
2. From nitriles and esters

(a) Stephen's Method

This reaction involves the preparation of aldehydes (R-CHO) from nitriles (R-CN) using tin(II) chloride ($SnCl_2$), hydrochloric acid (HCl).

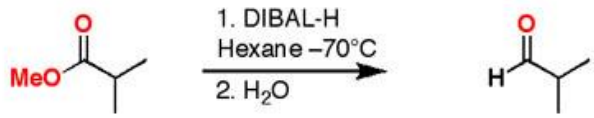
$$R - C \equiv N \xrightarrow[\text{HCl, 2. } H_2O]{\text{1. } SnCl_2} RCHO$$

Hydrolysis of intermediate (RCH = NH) gives RCHO and NH_3.



(b) From Esters

DIBAL (Di-isobutyl aluminum hydride) is a strong, bulky reducing agent. It's most useful for the reduction of alkyl cyanide or esters to aldehydes. Unlike lithium aluminum hydride, it will not reduce the aldehyde further if only one equivalent is added.



$$R - CN \xrightarrow{DIBAL-H} \xrightarrow{H_2O} R - CHO$$

Low temperature is important to prevent further reduction.

- **3. FROM HYDROCARBONS**

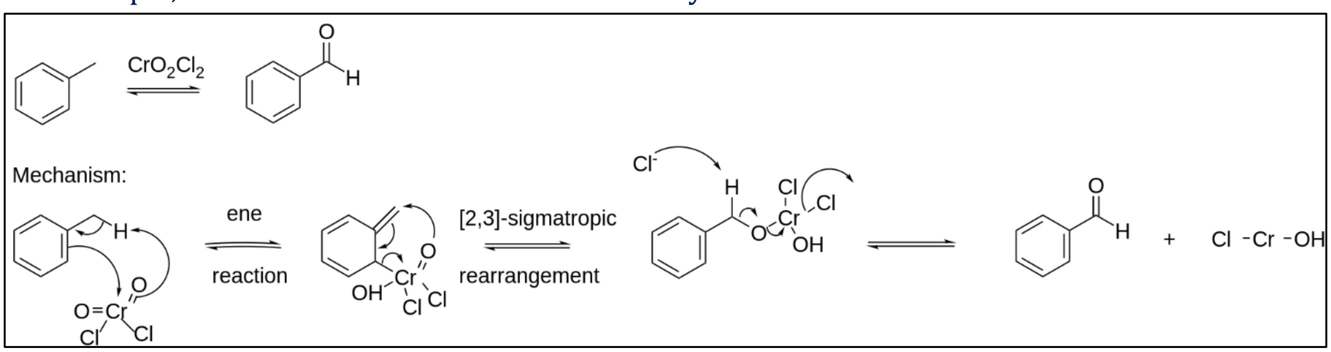
(i) BY THE OXIDATION OF METHYLBENZENE

Aldehydes are easily oxidised to carboxylic acids by the same reagent acidic dichromate, in their syntheses. Hence, removal of aldehyde as fast as it is formed is to be accomplished.

(a) Use of chromyl chloride [Etard reaction]

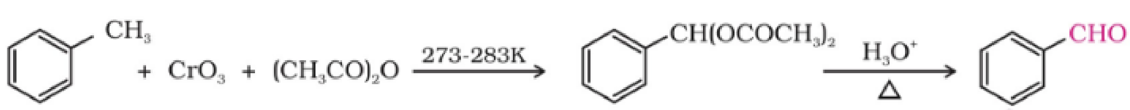
The Etard reaction is a chemical reaction that involves the direct oxidation of an aromatic or heterocyclic bound methyl group to an aldehyde using chromyl chloride.

For example, toluene can be oxidized to benzaldehyde.

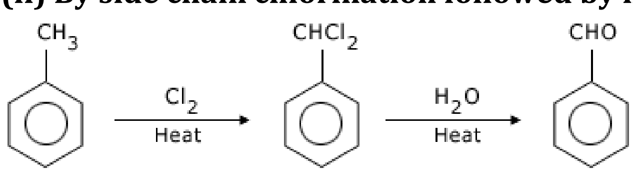


(b) Use of chromic oxide

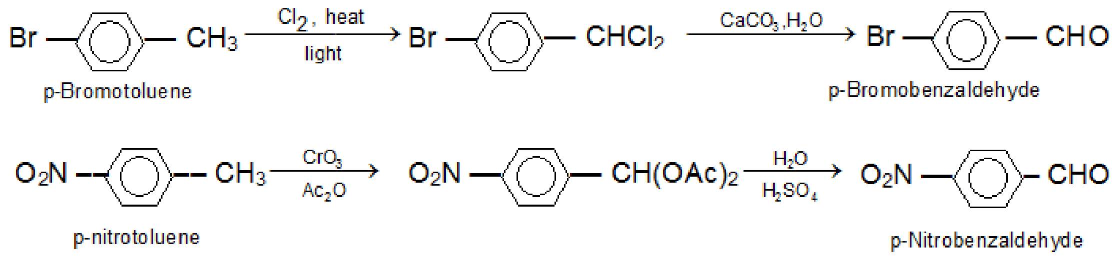
In the case of toluene, oxidation of the side chain can be interrupted by trapping the aldehyde in the form a non-oxidisable derivative, the gem-diacetate, which can be isolated and then hydrolysed.



(ii) By side chain chlorination followed by hydrolysis



Example.



(iii) Gatterman-Koch Aldehyde Synthesis:

Benzaldehyde may be synthesized by bubbling a mixture of carbon-monoxide and hydrogen chloride through a solution of either nitrobenzene or solution containing benzene and a catalyst of $AlCl_3$ and small amount of cuprous chloride.

$$C_6H_6 + CO + HCl \xrightarrow{AlCl_3} C_6H_5CHO + HCl$$

The mechanism of this reaction is uncertain, but it appears likely that the formyl cation is the active species.

$$CO + HCl + AlCl_3 \longrightarrow [CH = O \leftrightarrow H - C \equiv O^+] + AlCl_4^-$$

The Gatterman - Koch aldehyde synthesis is not applicable to phenols and their ethers, or when the substituent is strongly deactivating eg. nitrobenzene.

(iv) Gattermann Aldehyde Synthesis:

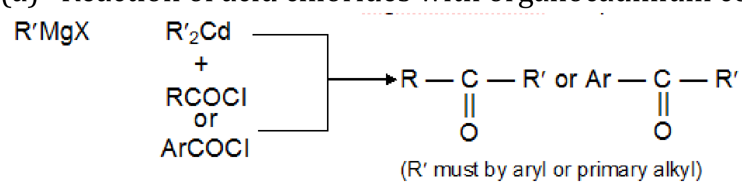
When benzene is treated with a mixture of HCN and HCl in the presence of $AlCl_3$, and the complex so produced decomposed with water, benzaldehyde is produced.

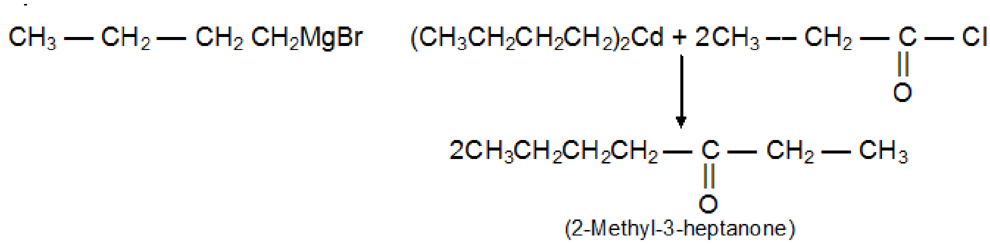
$$C_6H_6 + HCN + HCl \xrightarrow{AlCl_3} C_6H_5CH = NH.HCl \xrightarrow{H_2O} C_6H_5CHO + NH_3$$

- **PREPARATION OF KETONES**

1. With Organometallics

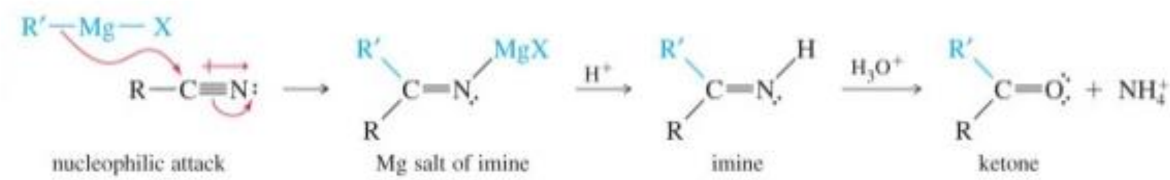
(a) Reaction of acid chlorides with organocadmium compounds





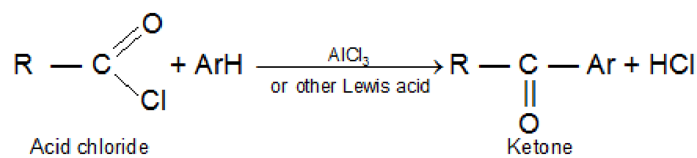
2. From nitriles

Nitriles, RC≡N, react with Grignard reagents or organolithium reagents to give ketones.



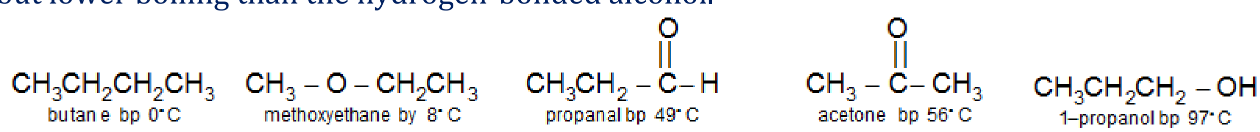
Nucleophiles can attack the electrophilic carbon of the nitrile group. Grignard (or organolithium) reagents attack the nitrile to generate the magnesium (or lithium) salt of an imine. Acid hydrolysis generates the imine, and under these acidic conditions, the imine is hydrolyzed to a ketone.

3. Friedel – Crafts acylation



Physical Properties of Aldehydes & Ketones

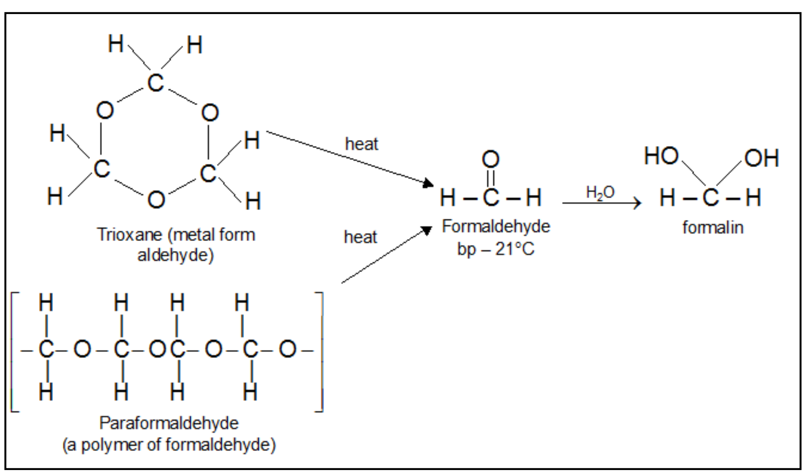
The ketones and the aldehyde are more polar and higher-boiling than the ether and the alkane, but lower boiling than the hydrogen-bonded alcohol.



Pure ketones and aldehydes cannot engage in hydrogen bonding with each other, the –OH of water or an alcohol can form a H– bond with the unshared electrons on a carbonyl oxygen atom. Because of this hydrogen bonding, ketones and aldehydes are soluble in water.

Acetaldehyde and acetone are miscible (soluble in all proportions) with water, and other ketones and aldehydes with up to four carbon atoms are appreciably soluble in water.

Formaldehyde is a gas at room temperature, so it is often stored and used as a 40 percent aqueous solution called formalin.



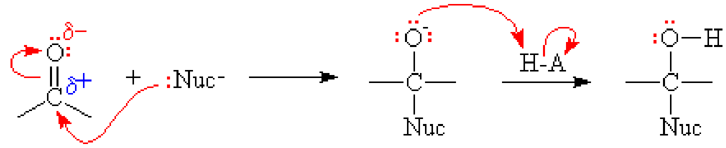
Acetaldehyde boils near room temperature, and it can be handled as a liquid.

Chemical Properties of Aldehydes & Ketones

- **1. Nucleophilic addition reactions**

(a) Mechanism

The C of the carbonyl group is electrophilic and initially forms a bond with the nucleophile and then the negatively charged oxygen in this intermediate tends to abstract a proton from the solvent to become an alcohol (with simultaneous conversion of the C-O back into C=O).



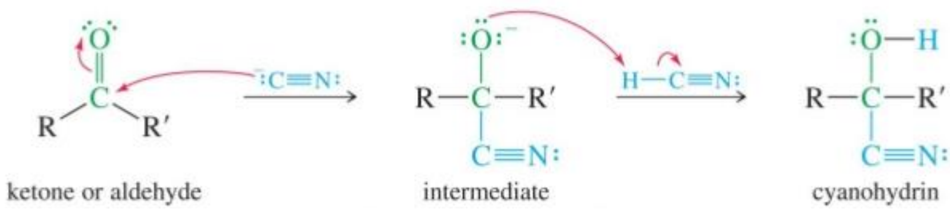
(b) Reactivity

Aldehydes are typically more reactive than ketones due to the following factors.

(i) Aldehydes are less sterically hindered than ketones.

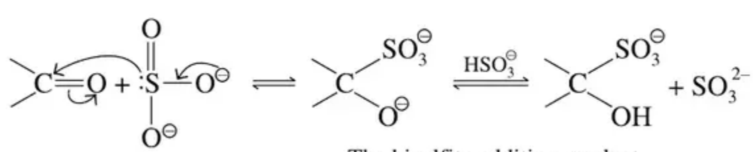
(ii) The carbonyl carbon in aldehydes generally has more partial positive charge than in ketones due to the electron-donating nature of alkyl groups.

(i) Addition of HCN



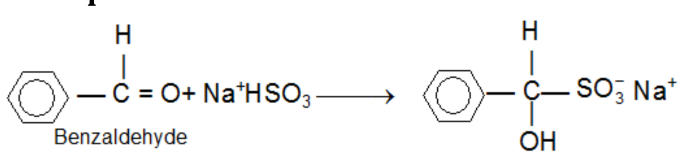
(ii) Addition of sodium hydrogensulphite

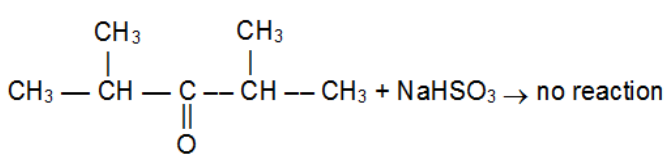
$$HSO_3^{\ominus} \rightleftharpoons H^{\oplus} + SO_3^{2-}$$



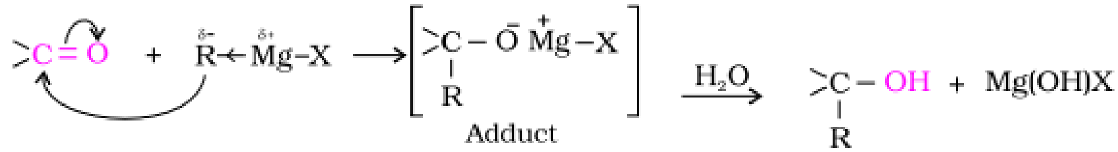
The bisulfite addition product

Example

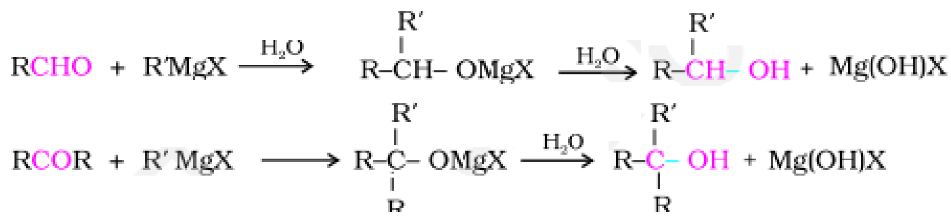




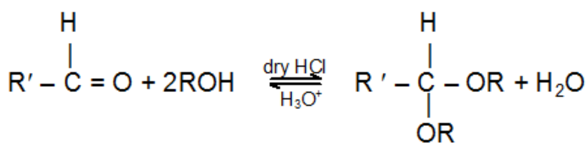
(iii) Addition of Grignard reagents



$$HCHO + RMgX \rightarrow RCH_2OMgX \xrightarrow{H_2O} RCH_2OH + Mg(OH)X$$



(iv) Addition of alcohols (Acetal Formation)



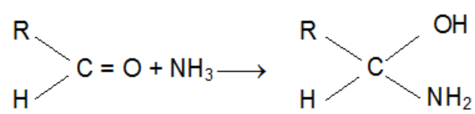
An acetal (gem-diether)

In H_3O^+, RCHO is regenerated because acetals undergo acid catalyzed cleavage much more easily than do ethers. Since acetals are stable in neutral or basic media, they are used to protect the –CH=O group.

(v) Addition of ammonia and its derivatives

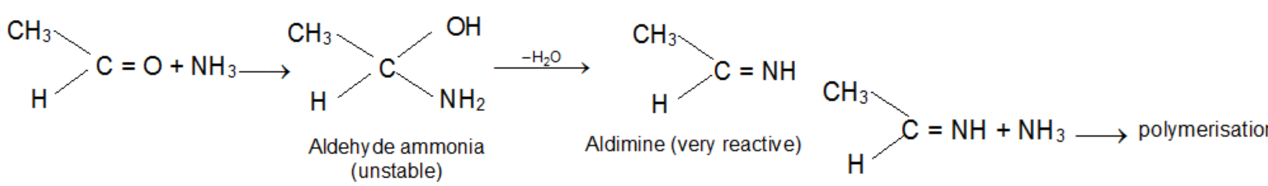
- **Addition of Ammonia**

Aldehydes react with ammonia to form aldehyde ammonia



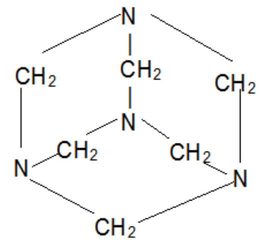
Aldehyde ammonia

The aldehyde ammonia is unstable and lose water immediately to form aldimine. The dehydration product is not usually obtained because, in most cases, it immediately polymerises to form cyclic trimers.



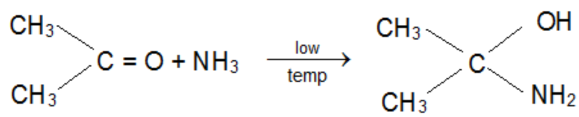
When treated with ammonia, formaldehyde does not form an aldehyde – ammonia, but gives instead hexamethylenetetramine, used in medicine as a urinary antiseptic under the name Urotoropine.

$$6HCHO + 4NH_3 \longrightarrow (CH_2)_6N_4 + 5H_2O \qquad (80\%)$$

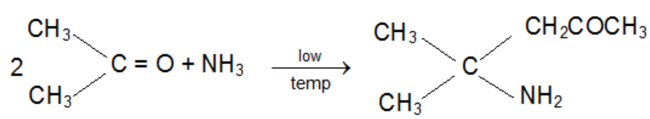


Hexamethylene tetramine

Ketones also give ketone-ammonia but these cannot be isolated. Acetone reacts slowly with ammonia to form acetone ammonia and then a complex compound.



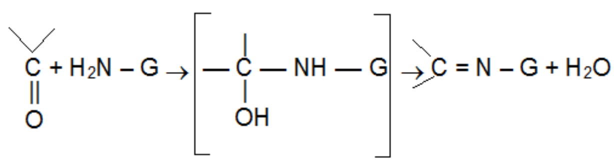
Acetone ammonia

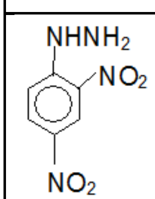
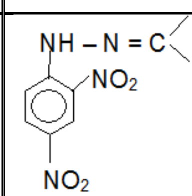


diacetone ammonia

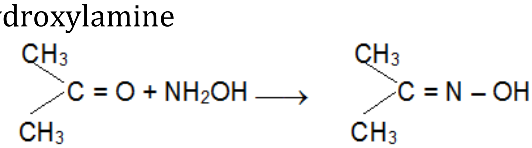
Aldimines, Schiff's bases or azomethines are formed when aldehydes react with aliphatic primary amines, which is removed by slow distillation.

- **Addition of Ammonia Derivatives**



	$H_2N – G$	**Product**	
$H_2N – NH_2$	Hydrazine	$> C = N – NH_2$	Hydrazone
$H_2N – NH – C_6H_5$	Phenylhydrazine	$> C = N – NHC_6H_5$	Phenylhydrazone
$H_2N– NH– CONH_2$	Semicarbazide	$> C = N –NHCONH_2$	Semicarbazone
	2, 4-Dinitrophenyl hydrazine		2, 4-dinitrophenylhydrazone (bright orange or yellow precipitate used for identifying aldehydes and ketones

e) Addition of hydroxylamine



- **2. Reduction**

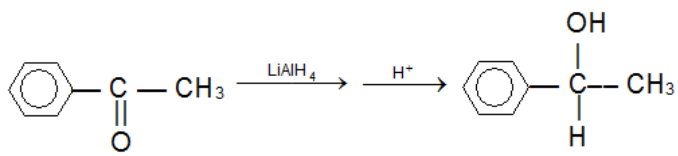
  (i) Reduction to alcohols

$$-\overset{|}{C}=O \xrightarrow[\text{LiAlH}_4\text{ or NaBH}_4;]{H_2 + \text{Pt or Pd}} -\underset{H}{\overset{|}{\underset{|}{C}}}-OH$$

Aldehydes → 1° alcohols; Ketones → 2°alcohols

Example:

$$CH_3-CH_2CHO \xrightarrow{H_2,Pt} CH_2CH_2CH_2OH$$
$$CH_3-CH_2CHO \xrightarrow[2.\ H^+]{1.\ LiAlH_4} CH_3CH_2CH_2OH$$



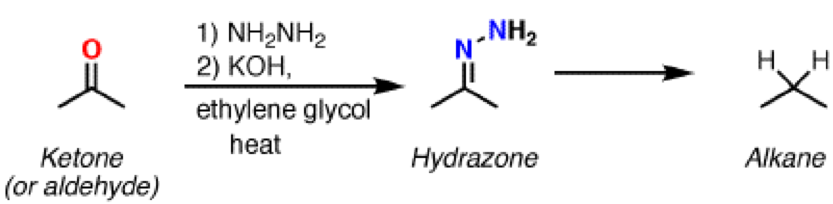
(ii) Reduction to hydrocarbons

(a) Clemmensen Reduction

$$CH_3CH_2-\overset{O}{\overset{||}{C}}-H \xrightarrow{Zn(Hg)\ /\ \text{conc. HCl}} CH_3CH_2CH_3$$

$$CH_3-\overset{O}{\overset{||}{C}}-CH_3 \xrightarrow{Zn(Hg)\ /\ \text{conc. HCl}} CH_3CH_2CH_3$$

(b) Wolff – Kishner Reduction



- **3. Oxidation**

$$\begin{array}{l} R\ CH=O \\ \text{or} \\ Ar-CH=O \end{array} \xrightarrow{KMnO_4\ \text{or}\ K_2Cr_2O_7H^+} \begin{array}{l} R-CO_2H \\ \text{or} \\ ArCO_2H \end{array}$$

Strong Oxidants: Ketones resist mild oxidation, but with strong oxidants at high temperature they undergo cleavage of C – C bonds (Carbonyl group goes with smaller alkyl gp.).

$$R-CH_2\overset{a}{-}\underset{\underset{O}{||}}{C}\overset{b}{-}CH_2R' \xrightarrow[\text{(drastic)}]{\text{oxidation}} \underbrace{RCO_2H + R'CH_2CO_2H}_{\text{From cleavage of bond (a)}} + \underbrace{RCH_2CO_2H + R'CO_2H}_{\text{From cleavage of bond (b)}}$$

(i) Tollen's test [Silver Mirror Test]

Tollen's Reagent A specific oxidant for RCHO is $[Ag(NH_3)_2]^+$

$$R-CHO + [Ag(NH_3)_2]^+ \longrightarrow R-COOH + Ag(\downarrow)$$

[Silver – mirror]

Tollen's test chiefly used for the detection of aldehydes.

Tollen's reagent does not attack carbon-carbon double bonds.

(ii) Fehling's test

Aldehydes can be oxidised by Fehling's solution.

Fehling's solution is made by mixing, Fehling A solution (which contains copper sulphate) and Fehling B solution [which contains sodium hydroxide and Rochelle salt (Sodium Potassium tartarate)].

During the oxidation of aldehydes to acids, the cupric ions are reduced to cuprous ions which are precipitated as red cuprous oxide.

$$RCHO + 2Cu^{+2} + 3OH^- \longrightarrow RCOO^- + 2Cu^+ + 2H_2O$$

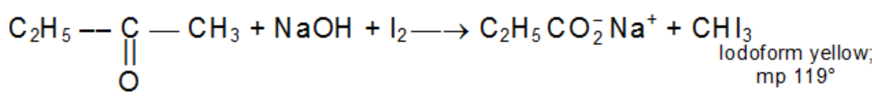
$$2Cu^+ + 2OH^- \longrightarrow Cu_2O + H_2O$$

Cuprous oxide (red)

(iii) Oxidation of methyl ketones (Haloform reaction)

CH_3COR are readily oxidised by NaOI ($NaOH + I_2$) to iodoform, CHI_3, and RCO_2Na.

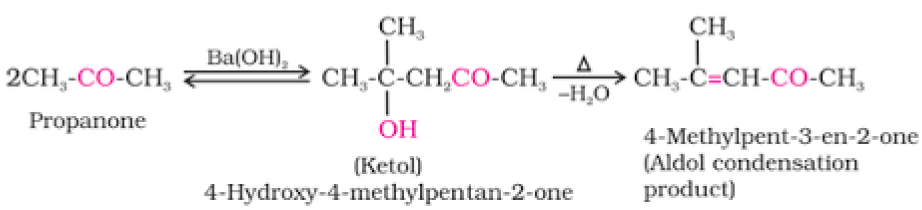
Example:



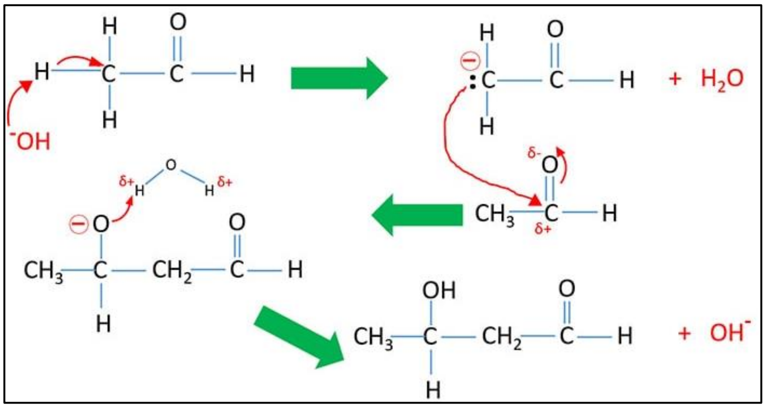
- **4. Reactions due to α– hydrogen**

(i) Aldol condensation

Under the influence of dilute base or dilute acid, two molecules of an aldehyde or ketone may combine to form a β-hydroxy aldehyde or ketone.



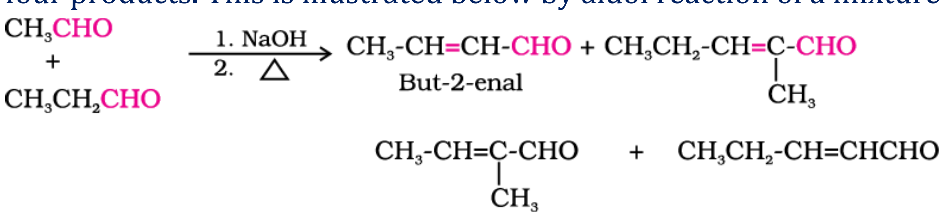
The α-carbanion generated by the base from one molecule of aldehyde or ketone adds to the carbonyl carbon of the other molecule and the two molecules condense and form β-hydroxy aldehyde or ketone.



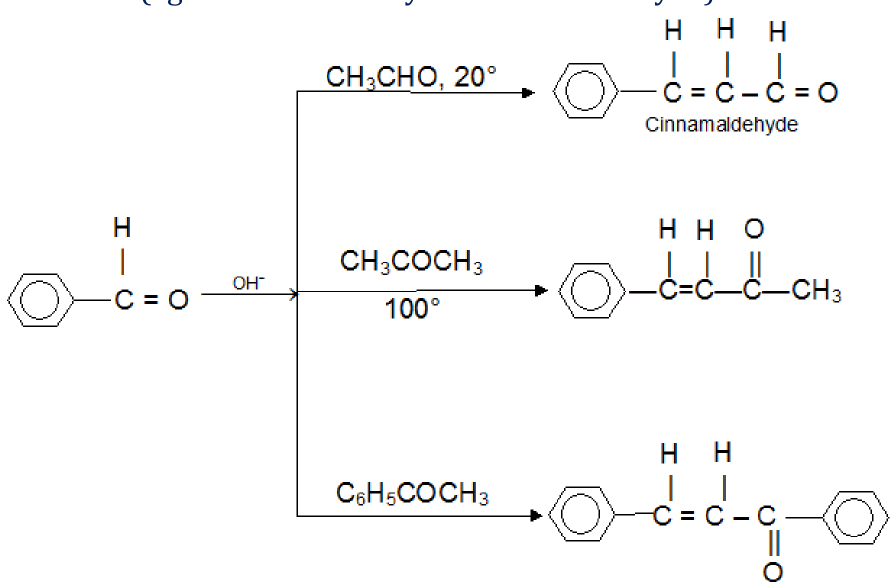
Aldol condensations are reversible, and with ketones the equilibrium is unfavourable for condensation product. β-hydroxycarbonyl compounds are readily dehydrated to give α,β unsaturated carbonyl compounds.

(ii) Cross aldol condensation

When aldol condensation is carried out between two different aldehydes and / or ketones, it is called cross aldol condensation. If both of them contain α-hydrogen atoms, it gives a mixture of four products. This is illustrated below by aldol reaction of a mixture of ethanal and propanal



Under certain condition, a good yield of a single product can be obtained from a crossed aldol condensation. One reactant contains no α-hydrogens and therefore is incapable of condensing with itself (eg. Aromatic aldehydes or formaldehyde).



- **5. Other reactions**

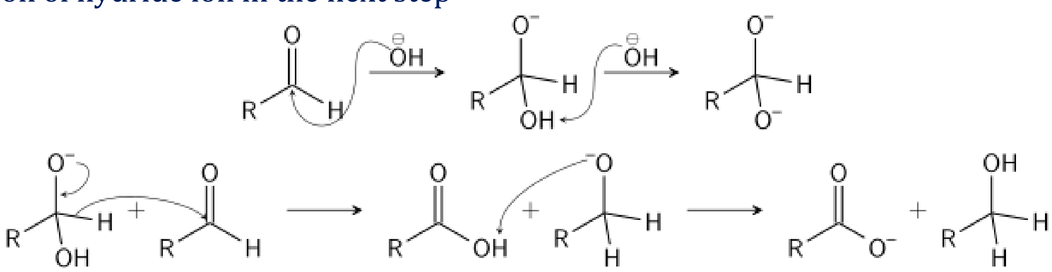
(i) Cannizzaro reaction (Disproportionation Reaction)

In the presence of concentrated alkali, aldehydes containing no – α hydrogens undergo self Oxidation and reduction to yield a mixture of an alcohol and a salt of a carboxylic acid. This reaction is known as **Cannizzaro – reaction**.

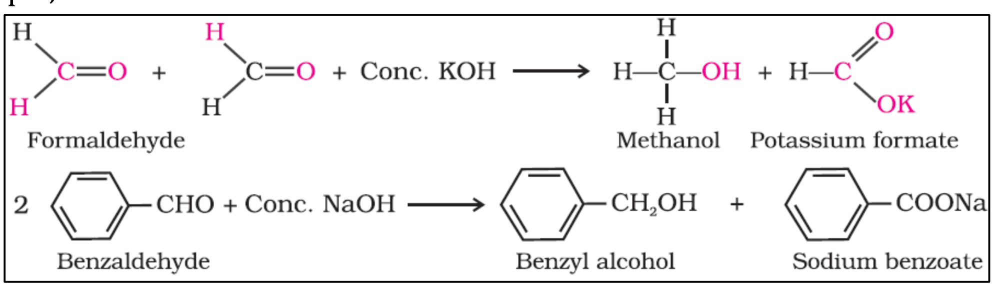
Two successive additions are involved.

a) Addition by hydroxide ion in first step

b) Addition of hydride ion in the next step



For example,



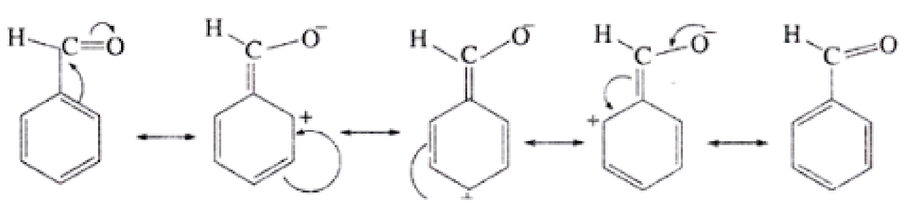
This explains the Crossed Cannizzaro reaction, in which two aldehydes (with no α– H atom) undergo disproportionation reaction to form alcohol and carboxylic acid.

$$C_6H_5CHO + HCHO \xrightarrow{\text{50\% NaOH}} C_6H_5CH_2OH + HCOO^-Na^+$$

(ii) Electrophilic Aromatic Substitution Reactions of Benzaldehyde

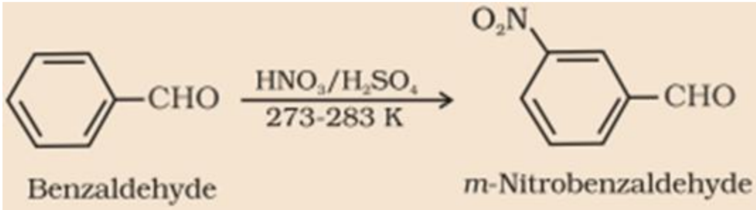
The —CHO group is strongly deactivating and m– directing.

Benzaldehyde does not give Friedel – Craft alkylation or acylation reactions.



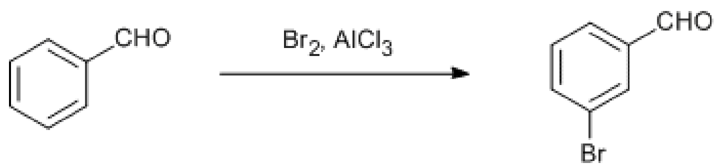
(a) Nitration of Benzaldehyde:

At 273 – 283 K with fuming nitric acid, in the presence of sulphuric acid, benzaldehyde yields m–nitrobenzaldehyde.



(b) Halogenation of Benzaldehyde:

On treating benzaldehyde with Br_2, in the presence of anhydrous $AlCl_3$, m–bromobenzaldehyde formed.



- **Uses of Aldehydes and Ketones**

  (i) Uses of Aldehydes: Aldehyde is mostly used in the formation of resins, when it is combined with melamine, urea, phenol etc. butyraldehyde is mainly used as a plasticizer. Some other aldehyde is used as ingredients in flavours and deodorants. Other aldehydes have commercial uses also like oxo-alcohol, which is used in detergent.

  (ii) Applications of Ketone: Ketones are produced at very high scale as pharmaceuticals, solvents, polymers in industries. Acetone, cyclohexanone, methyl ethyl ketone are amongst the most important ketones.

- **Analysis of Aldehydes and Ketones**

  Aldehydes and ketones are characterized through the addition to the carbonyl group of nucleophilic reagents, especially derivatives of ammonia. Aldehydes and ketones are generally identified through the melting points of derivatives like 2,4-dinitrophenylhydrazones, oximes, and semicarbazones.

  All aldehyde or ketone will, for example react with 2,4-dinitrophenylhydrazine to form an insoluble yellow or red solid.

  Aldehydes are characterized, and in particular are differentiated from ketones through their ease of oxidation: aldehydes give a positive test with Tollen's reagent; ketones do not.

  Aldehydes can be oxidised by Fehling's solution.

  Aldehydes are also, of course, oxidized by many other oxidizing agents : by cold, dilute, neutral $KMnO_4$ and by CrO_3 in H_2SO_4.

  A highly sensitive test for aldehydes is the Schiff's test.

  Methyl Ketones are characterized through the iodoform test.

Carboxylic Acids and Its Derivatives

Carboxylic acids contain the carboxyl group attached to hydrogen (HCOOH) an alkyl group (RCOOH), or an aryl group (ArCOOH). The general formula of the carboxylic acids is $C_nH_{2n}O_2$.

Nomenclature of carboxylic acids

According to the IUPAC system of nomenclature, the suffix of the monocarboxylic acid is 'oic acid', which is added to the name of the alkane corresponding to the longest carbon chain containing the carboxyl group, e.g.,

HCOOH	Methanoic acid
$CH_3-CH_2-CH_2-COOH$	Butanoic acid

The positions of side-chains (or substituents) are indicated by numbers, the carboxyl group always being given number I.

$$CH_3-\underset{CH_3}{\underset{|}{CH}}-\underset{CH_3}{\underset{|}{CH}}-CH_2-COOH$$

3,4–dimethylpentanoic acid

Naming of Acyl Groups; Acid halides, acid anhydrides, esters and acid amides:

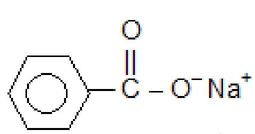
The group obtained from a carboxylic acid by the removal of the hydroxyl portion is known as an acyl group. The name of an acyl group is created by changing the 'ic' at the end of the name of the carboxylic acid to 'yl', examples:

(i) Acid chlorides are named systematically as acyl chlorides.

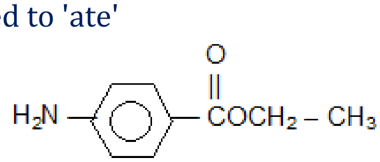
$$CH_3-\overset{\overset{O}{||}}{C}-Cl$$ Ethanoylchloride

(ii) An acid anhydride is named by substituting anhydride for acid in the name of the acid from which it is derived.

(iii) The name of the cation (in the case of a salt) or the name of the organic group attached to the oxygen or the carboxyl group (in the case of an ester) preceeds the name of the acid. The 'ic acid' part of the name of the acid is converted to 'ate'



Sodium benzoate



Ethyl-p-aminobenzoate

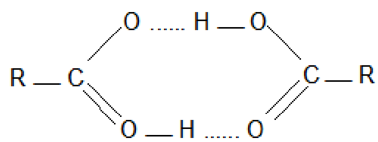
(iv) The names of amides are formed by replacing –oic acid (or –ic acid for common names) by amide or –carboxylic acid by carboxamide.

$$CH_3-\overset{\overset{O}{||}}{C}-NH_2$$ Ethanamide or acetamide

Physical Properties of Carboxylic Acids:

The molecules of carboxylic acids are polar and exhibit hydrogen bonding. The first four are miscible with water. The higher acids are virtually insoluble. The simplest aromatic acid, benzoic acid, contains too many carbon atoms to show appreciable solubility in water.

Carboxylic acids are soluble in less polar solvents like ether, alcohol, benzene etc



Carboxylic acids have higher boiling points than alcohols. These very high boiling points are due to the fact that a pair of carboxylic acid molecules are held together not by one but by two hydrogen bonds and exist as dimer.

The first three fatty acids are colourless pungent smelling liquids.

General Methods of Preparation of Carboxylic Acids

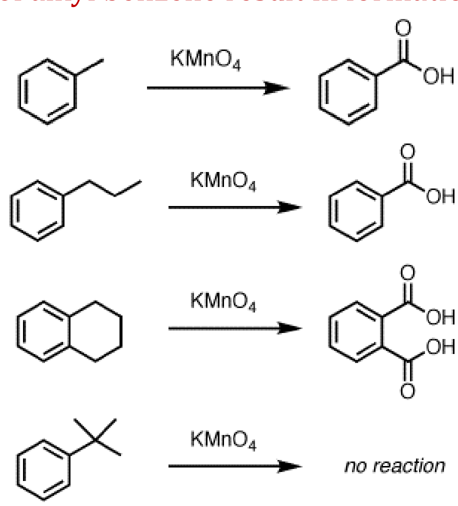
1. Oxidation of Primary Alcohols & Aldehydes

Primary alcohols on oxidation give aldehydes, the reaction does not stop here and the oxidation continues to give carboxylic acids as final product.

$$R-CH_2OH \xrightarrow[\text{[O]}]{K_2Cr_2O_7 + H_2SO_4} RCHO \xrightarrow[\text{[O]}]{K_2Cr_2O_7 + H_2SO_4} RCOOH$$

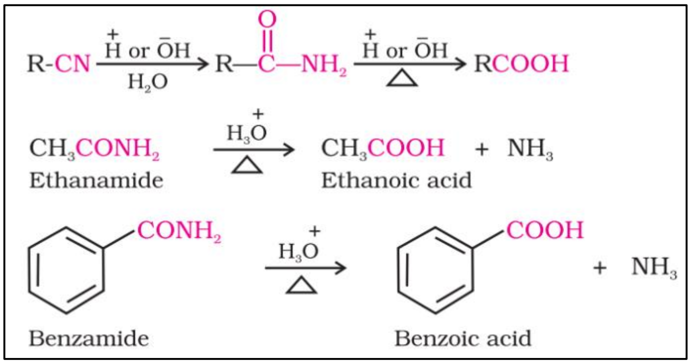
2. Oxidation of Alkyl Benzenes

Strong oxidation of alkyl benzene result in formation of carboxylic acids.



3. From Nitriles (Cyanides) and amides:

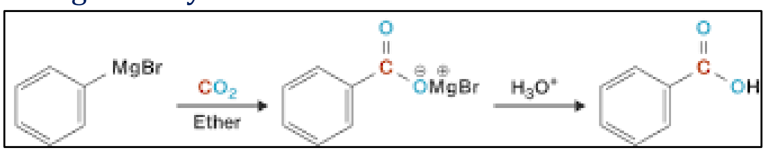
Nitriles are hydrolysed to amides and then to acids in the presence of H^+ or OH^- as catalyst. Mild reaction conditions are used to stop the reaction at the amide stage.



This synthetic method is generally limited to the use of primary alkyl halides. Aryl halides (except for those with ortho and para nitro groups) do not react with sodium cyanide.

4. Grignard Synthesis of Carboxylic Acid

Grignard reagents react with carbon dioxide (dry ice) to form salts of carboxylic acids which in turn give corresponding carboxylic acids after acidification with mineral acid.



This method is useful for converting alkyl halides into corresponding carboxylic acids having one carbon atom more than that present in alkyl halides (ascending the series)

5. Hydrolysis of acyl halides and anhydrides

Acid chlorides on hydrolysis give carboxylic acids.

$$R - COCl \xrightarrow{H_2O} R - COOH + Cl^-$$

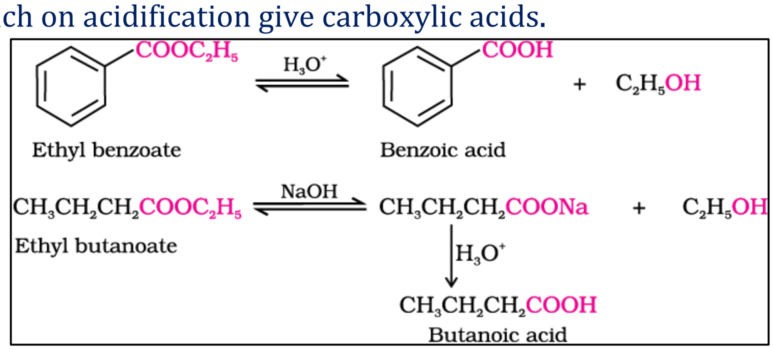
Acid Anhydrides on the other hand give corresponding acid on hydrolysis.

$$\underset{\text{Benzoic anhydride}}{(C_6H_5CO)_2O} \xrightarrow{H_2O} \underset{\text{Benzoic acid}}{2\ C_6H_5COOH}$$

$$\underset{\text{Benzoic ethanoic anhydride}}{C_6H_5COOCOCH_3} \xrightarrow{H_2O} \underset{\text{Benzoic acid}}{C_6H_5COOH} + \underset{\text{Ethanoic acid}}{CH_3COOH}$$

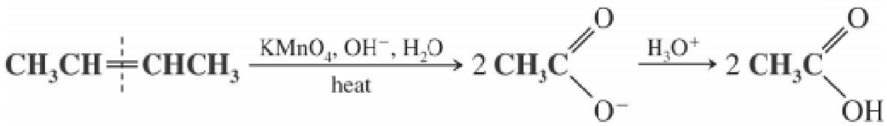
6. Hydrolysis of esters

Acidic hydrolysis of esters gives directly carboxylic acids while basic hydrolysis gives carboxylates, which on acidification give carboxylic acids.



7. Oxidation of Alkenes and alkynes:

Strong oxidation of alkenes results in formation of carboxylic acids. Alkenes can be oxidized to carboxylic acids with hot alkaline $KMnO_4$.



(cis or trans) Acetate ion Acetic acid

$$R-C\equiv C-R' \xrightarrow[(2)\ H_3O^+]{(1)\ KMnO_4,\ OH^-} RCO_2H + R'CO_2H$$

8. Hydrolysis of Trihalogen Derivatives

Trihalogen derivatives in which the three halogen atoms are all attached to the same carbon atom yields carboxylic acid on hydrolysis.

$$RCX_3 \xrightarrow{KOH} R-C(OH)_3 \longrightarrow RCO_2H + H_2O$$

9. Oxidation of Methyl Ketone

Methyl ketone can be converted to carboxylic acids via the haloform reaction.

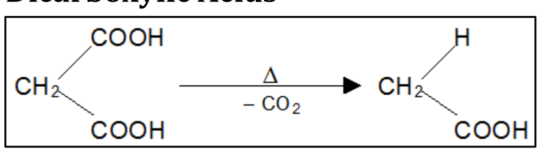
$$Ar-\overset{\overset{\Large O}{\|}}{C}-CH_3 \xrightarrow[2.\ H_3O^+]{1.\ X_2/NaOH} Ar-\overset{\overset{\Large O}{\|}}{C}-OH + CHX_3$$

10. Koch Reaction

A recent method for manufacturing fatty acids is to heat an olefin with carbon monoxide and steam under pressure at 300-400° C in the presence of a catalyst, e.g. phosphoric acid.

$$CH_2 = CH_2 + CO + H_2O \rightarrow CH_3 - CH_2 - COOH$$

11. Heating Gem Dicarboxylic Acids



Chemical Properties of Carboxylic Acids

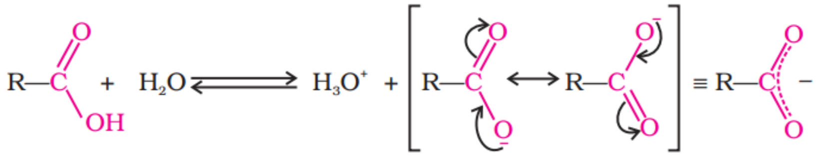
1. Acidity of Carboxylic Acids

Carboxylic acids are weak acids and their carboxylic anions are strong conjugate bases are slightly alkaline due to the hydrolysis of carboxylate anion compared to other species, the order of acidity and basicity or corresponding conjugate bases are as follows:

Acidity $RCOOH > HOH > ROH > HC\equiv CH > NH_3 > RH$

Basicity $RCOO^- < HO^- < RO^- < HCC^- < NH_2^- < R^-$

Carboxylic acids dissociate in water to give resonance stabilised carboxylate anions and hydronium ion.



Because of the resonance, both the carbon oxygen bond in the carboxylate anion have identical bond length. In the carboxylic acid, these bond lengths are no longer identical.

- **Effect of substituents on the acidity of carboxylic acids**

Electron withdrawing group (EWG) stabilises the carboxylate anion and strengthens the acid

Electron donating group (EDG) destabilises the carboxylate anion and weakens the acid

The acidity of carboxylic acid depends very much on the substituent attached to – COOH group. Electron withdrawing groups disperse the negative charge and thus stabilize the anion which results in increase in acidity of the carboxylic acids. Electron donating groups intensify the negative charge and the destabilize the anion which results in decrease in acidity of carboxylic acid. For example:

a) Increase in the number of Halogen atoms on a-position increases the acidity, eg.

$$Cl_3CCOOH > Cl_2CHCOOH > ClCH_2COOH > CH_3COOH$$

b) Increase in the distance of Halogen from COOH decreases the acidity e.g.

$$CH_3 - CH_2 - CH(Cl) - COOH > CH_3 - CH(Cl) - CH_2 - COOH > ClCH_2 - CH_2 - CH_2 - COOH$$

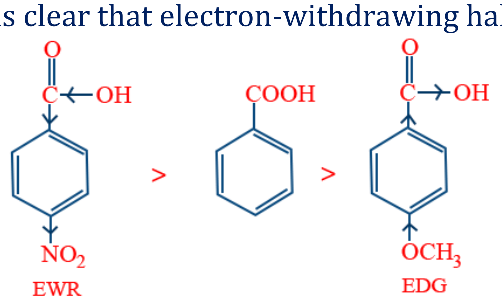
This is due to the fact that inductive effect decreases with increasing distance.

c) Increase in the electro negativity of halogen increases the acidity.

$$FCH_2COOH > BrCH_2COOH > ICH_2COOH$$

Carboxylic Acids	**K_a Value**
HCOOH	17.7×10^{-5}
CH_3COOH	1.75×10^{-5}
$ClCH_2COOH$	136×10^{-5}
$Cl_2CHCOOH$	5530×10^{-5}
Cl_3CCOOH	23200×10^{-5}

From the table it is clear that electron-withdrawing halogen increase acidity of carboxylic acids.



2. Reaction of Carboxylic Acids with Metals

The carboxylic acids react with metals to liberate hydrogen and are soluble in both NaOH and $NaHCO_3$ solutions. For example,

$$2CH_3COOH + 2Na \rightarrow 2CH_3COO^-Na^+ + H_2$$

$$CH_3COOH + NaOH \rightarrow CH_3COO^-Na^+ + H_2O$$

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COO^-Na^+ + H_2O + CO_2$$

3. Conversion of carboxylic acids into functional derivatives

Carboxylic acids can be converted into number of other compounds (known as derivatives of carboxylic acids or simply acid derivatives) by replacement of its –OH group by a Cl, OR or NH_2.

- Replacement of -OH by -Cl forms acid chlorides.
- Replacement of -OH by -OOCR forms acid anhydride.
- Replacement of -OH by -OR forms ester.
- Replacement of -OH by $-NH_2$ forms amide.

Carboxylic acid can be recovered simply by hydrolysis of the acid derivatives or one can say that the functional derivatives are all readily reconverted into the acid by simple hydrolysis.

(i) Conversion of carboxylic acids into acid halide (Reaction of carboxylic acids with PCl_5, PCl_3 and $SOCl_2$)

The hydroxyl group of carboxylic acids is replaced by chlorine atom on treating with PCl_5, PCl_3 or $SOCl_2$.

$$RCOOH + PCl_5 \rightarrow RCOCl + POCl_3 + HCl$$

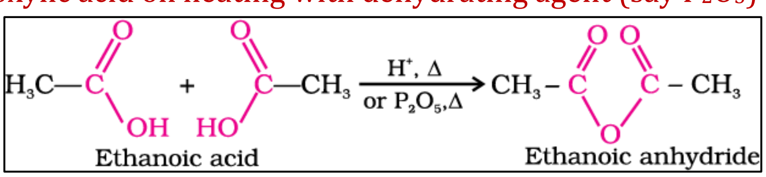
$$RCOOH + PCl_3 \rightarrow 3RCOCl + H_3PO_3$$

$$RCOOH + SOCl_2 \rightarrow RCOCl + SO_2 + HCl$$

Thionyl chloride ($SOCl_2$) for this reaction because the other two products are gaseous and escape the reaction mixture and we get a pure product.

(ii) Conversion of carboxylic acids into acid anhydrides

Lower monocarboxylic acid on heating with dehydrating agent (say P_2O_5) forms anhydrides.



Note: Anhydride of formic acid is not known, it gives CO and H_2O on heating with conc. H_2SO_4.

$HCOOH + \text{Conc. } H_2SO \rightarrow H_2O + CO$

(iii) Conversion of carboxylic acids into Esters (Esterification)

Carboxylic acids are esterified with alcohols or phenols in the presence of a mineral acid such as concentrated H_2SO_4 or HCl gas as a catalyst.

$$RCOOH + R'OH \underset{}{\overset{H^+}{\rightleftharpoons}} RCOOR' + H_2O$$

This reaction is reversible and the same catalyst, hydrogen ion, that catalyzes the forward reaction i.e. esterification necessarily catalyzes the reverse reaction i.e. hydrolysis.

The equilibrium is particularly unfavorable when phenols (ArOH) are used instead of alcohol; yet if water is removed during the reaction, phenolic esters (RCOOAr] are obtained in high yield.

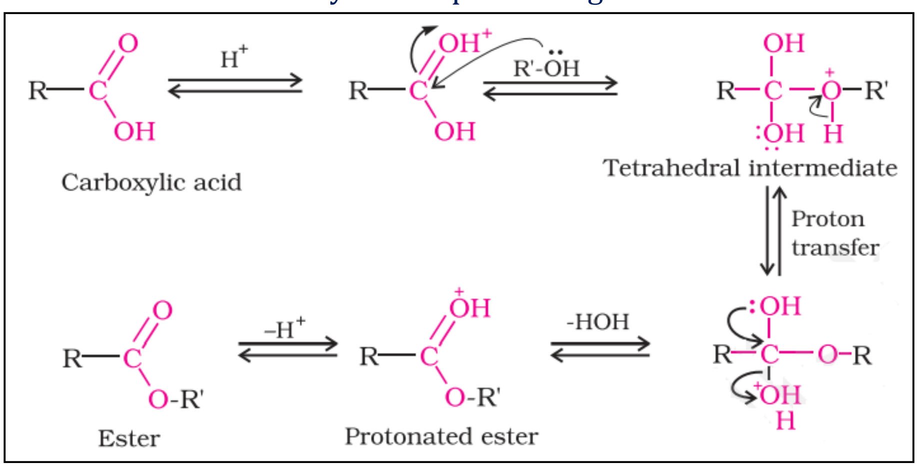
The presence of bulky group near the site of reaction, whether in alcohol or in the acid, slows down esterification (as well as its reverse, hydrolysis).

Reactivity $CH_3OH > 1° > 2° > 3°$

In esterification $HCOOH > CH_3COOH > RCH_2COOH > R_2CHCOOH > R_3CCOOH$

- **Mechanism of Esterification**

The esterification of carboxylic acids with alcohols is a kind of nucleophilic acyl substitution. Protonation of the carbonyl oxygen activates the carbonyl group towards nucleophilic addition of the alcohol. Proton transfer in the tetrahedral intermediate converts the hydroxyl group into $-{}^{+}OH_2$ group, which, being a better leaving group, is eliminated as neutral water molecule. The protonated ester so formed finally loses a proton to give the ester

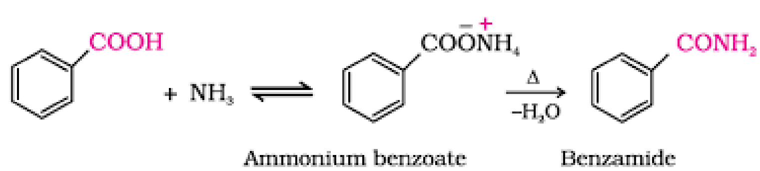


A carboxylic acid does not react with an alcohol unless a strong acid is used as a catalyst, protonation makes the carbonyl group more electrophilic and enables it to react with the alcohol, which is a weak nucleophile.

(iv) Conversion of carboxylic acids into amides:

Reaction of carboxylic acids with ammonia results in formation of give ammonium salt which on further heating at high temperature give amides.

$$\underset{}{CH_3COOH} + NH_3 \rightleftharpoons \underset{\text{Ammonium acetate}}{CH_3\overset{-}{CO}\overset{+}{ONH_4}} \xrightarrow[-H_2O]{\Delta} \underset{\text{Acetamide}}{CH_3CONH_2}$$



5. Reduction of carboxylic acids to alcohols:

Lithium aluminium hydride can reduce an acid to an alcohol; the initial product is an alkoxide from which the alcohol is liberated by hydrolysis.

$4\ R\text{–}COOH + 3LiAlH_4 \rightarrow 4H_2 + 2LiAlO_2 + (RCH_2O)_4\ AlLi + H_2$

$(RCH_2O)_4\ AlLi + 4H_2O \rightarrow 4RCH_2OH + Al(OH)_3 + LiOH$

6. Decarboxylation (The process of removal of CO_2)

Saturated monocarboxylic acid salt of sodium or potassium on dry distillation with soda lime gives alkane.

$$RCOONa + NaOH \xrightarrow{[CaO],\ \Delta} R-H + Na_2CO_3$$

For example,

(I) The decarboxylation of sodium formate yields H_2.

$$HCOONa + NaOH \xrightarrow{[CaO],\ \Delta} H-H + Na_2CO_3$$

(II) The decarboxylation of sodium acetate yields CH_4.

$$CH_3COONa + NaOH \xrightarrow{[CaO],\ \Delta} CH_4 + Na_2CO_3$$

(i) The alkane formed by decarboxylation always contains one less than the original acid.

(ii) The yield is good in case of lower members but poor for higher members.

(iii) Soda lime is [NaOH + CaO]. CaO used alongwith NaOH keeps it dry (NaOH is hygroscopic).

7. Kolbe's electrolysis method:

Electrolysis of aqueous solution of sodium or potassium salt of saturated mono carboxylic acid forms alkane (at anode).

$$2RCOOK(aq) + 2H_2O \xrightarrow{\text{Electrolysis}} \underset{\text{At Anode}}{R-R(g) + 2CO_2(g)} + \underset{\text{At Cathode}}{H_2(g) + 2KOH(aq)}$$

Electrolysis of an acid salt gives symmetrical alkane. However, in case of mixture of carboxylic acid salts, all probable alkanes are formed.

$$RCOOK + R'COOK + 2H_2O \xrightarrow{\text{Electrolysis}} [R-R + R-R' + R'-R'] + 2CO_2 + H_2 + 2KOH$$

8. Halogenation of aliphatic carboxylic acids (Hell-Volhard-Zelinsky Reaction)

In the presence of phosphorus, aliphatic carboxylic acids react smoothly with chlorine or bromine to yield a compound in which α-hydrogen has been replaced by halogen.

$$CH_3-COOH \xrightarrow{Cl_2\ /P} Cl-CH_2-COOH \xrightarrow{Cl_2/P} Cl_2CH-COOH \xrightarrow{Cl_2/\ P} Cl_3CCOOH$$

The function of the phosphorus is ultimately to convert a little of the acid into acid halide so it is the acid halide, not the acid itself, that undergoes this reaction.

$$P + X_2 \longrightarrow PX_3$$

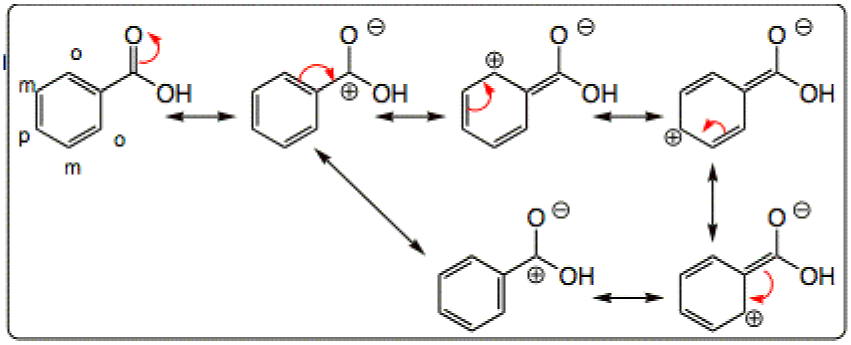
$$R-CH_2-COOH + PX_3 \longrightarrow RCH_2-COX$$

$$R-CH_2-COX + X_2 \longrightarrow \underset{\displaystyle X}{\underset{|}{R-CH}}-COX + HX \xrightarrow{H_2O} \underset{\displaystyle X}{\underset{|}{R-CH}}-CO_2H$$

9. Electrophilic Aromatic Substitution Reactions of Benzoic acid

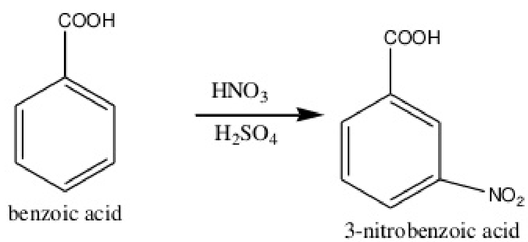
The —COOH group is strongly deactivating and m– directing.

Benzoic acid does not give Friedel – Craft alkylation or acylation reactions.



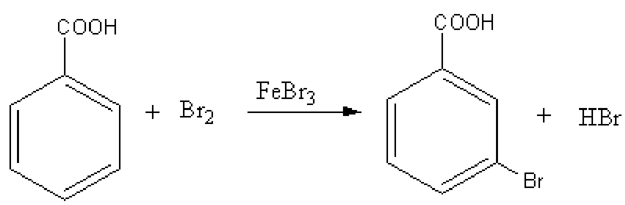
(a) Nitration of Benzaldehyde:

Benzoic acid reacts with fuming nitric acid, in the presence of sulphuric acid, yields m–nitrobenzoic acid.



(b) Halogenation of Benzaldehyde:

On treating benzoic acid with Br_2, in the presence of anhydrous $FeBr_3$, m–bromobenzoic acid formed.



Uses of Carboxylic Acids:

Methanoic acid is used in rubber, textile, dyeing, leather and electroplating industries. Ethanoic acid is used as solvent and as vinegar in food industry. Hexanedioic acid is used in the manufacture of nylon-6, 6. Esters of benzoic acid are used in perfumery. Sodium benzoate is used as a food preservative. Higher fatty acids are used for the manufacture of soaps and detergents.

THE END